Discrete-Time Markov Chains: 20 Worked Problems

Matrices & P^n · State Classification · Periodicity/Ergodicity · Stationary $\pi = \pi P$ · Quant Capstones

A Markov chain is a machine with one rule: *the next step depends only on where you are now.* That single rule turns an infinite history into a small table — the transition matrix — and everything in this chapter is a question about that table: where do you land after n steps (P^n), which states can you never come back to (classification), does the chain have a rhythm (periodicity), and where does it settle ($\pi = \pi P$)? Each problem is color-coded and carries an analogy. Several use the *same* chains so you can watch one table answer many questions.

A. Transition Matrices, Diagrams & P^n

Problem A1 — Building P from words

Each day is Sunny (S) or Rainy (R). After a sunny day, tomorrow is sunny with probability 0.8; after a rainy day, tomorrow is sunny with probability 0.4. Write P and draw the transition diagram.

Solution. Row = "from", column = "to". Each row must sum to 1:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \begin{matrix} \text{(from S)} \\ \text{(from R)} \end{matrix}$$



★ *Intuition:* Each row is a "menu of tomorrows": a fixed set of probabilities for where you go next, summing to 1 like slices of a pie. The diagram is the same table drawn as a subway map — stations are states, arrows are one-day rides.

Problem A2 — Two steps via Chapman–Kolmogorov

For A1's chain, find $P(\text{sunny in two days} \mid \text{sunny today})$.

Solution. Sum over tomorrow's weather (the intermediate state):

$$P^2_{SS} = P_{SS}P_{SS} + P_{SR}P_{RS} = 0.8(0.8) + 0.2(0.4) = \boxed{0.72}.$$

Full square: $P^2 = \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}$.

★ *Intuition:* To get from S to S in two hops you must stop somewhere in the middle — at S or at R. Add up every "via" route. P^2 is just that bookkeeping done for all pairs at once: matrix multiplication *is* route-counting.

Problem A3 — Evolving a whole distribution

It is rainy today. Find the distribution of tomorrow's, the day after's, and the third day's weather.

Solution. Start $\mu_0 = (0, 1)$ and multiply by P each day ($\mu_{n+1} = \mu_n P$):

$$\mu_1 = (0.4,\ 0.6), \qquad \mu_2 = (0.56,\ 0.44),$$

$$\mu_3 = (0.56(0.8) + 0.44(0.4),\ \ldots) = (0.624,\ 0.376).$$

So $P(\text{sunny on day 3} \mid \text{rain today}) = \boxed{0.624}$.

★ *Intuition:* A drop of dye (all probability on R) spreads through the weather states. Each multiplication by P is one tick of the clock; the dye concentration drifts toward a fixed pattern (Problem D1).

Problem A4 — A three-regime market

A market is Bull, Flat or Bear with

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

Find $P(\text{Bull in two days} \mid \text{Bear today})$.

Solution. Row "Bear" times column "Bull":

$$P^2_{\text{Bear,Bull}} = 0.2(0.7) + 0.3(0.3) + 0.5(0.2)$$

$$= 0.14 + 0.09 + 0.10 = \boxed{0.33}.$$

★ *Intuition:* Three possible "layovers" (Bull, Flat, Bear) on the way from Bear to Bull. Multiply the two legs of each itinerary, add the three itineraries. You never need to compute all of P^2 — one row times one column is enough.

▶ *Speed trick:* For one entry of P^2, do a dot product of one row with one column. Do the full matrix only if you need it.

Problem A5 — Probability of a specific path

In A4's chain, starting from Flat, find $P(X_1 = \text{Bull}, X_2 = \text{Bear}, X_3 = \text{Flat} \mid X_0 = \text{Flat})$.

Solution. By the Markov property the path probability is a product of one-step probabilities:

$$P_{F\to B} \cdot P_{B\to \text{Bear}} \cdot P_{\text{Bear}\to F} = 0.3 \times 0.1 \times 0.3 = \boxed{0.009}.$$

★ *Intuition:* The chain has no memory beyond the current state, so a trip's likelihood is just the product of its legs — like a multi-flight itinerary where each connection is independent of how you arrived at the airport.

B. State Classification

Problem B1 — Absorbing states and communicating classes

States $1, 2, 3, 4$ with

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0.3 & 0 \\ 0 & 0.2 & 0.5 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Classify the states, and find the probability that a chain started at state 2 is eventually absorbed in state 1.

Solution. States 1 and 4 are **absorbing** ($P_{ii} = 1$). States 2, 3 communicate with each other but can leak into 1 or 4 and never return: the class $\{2, 3\}$ is **transient**. First-step analysis with $h_i = P(\text{absorbed in 1} \mid \text{start } i)$:

$$h_2 = 0.3 + 0.4h_2 + 0.3h_3, \qquad h_3 = 0.2h_2 + 0.5h_3.$$

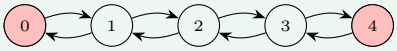
So $h_3 = 0.4h_2$ and $h_2(1 - 0.4 - 0.12) = 0.3 \Rightarrow h_2 = \boxed{0.625 = \frac{5}{8}}$.

★ *Intuition:* A casino floor with two exit doors (1 and 4). Once through a door you never come back. The tables inside (2, 3) are transient: you may linger, but you will leave eventually. Which door you take is decided by where you start — state 3 is nearer door 4.

Problem B2 — Gambler's ruin chain

A fair-coin gambler holds \$1–\$3 and stops at \$0 or \$4. Classify the states and find $P(\text{reaching \$4} \mid \text{start at \$1})$.

Solution.



States 0 and 4 are absorbing; $\{1, 2, 3\}$ is transient (positive

chance of hitting a boundary and never returning). For a fair walk the absorption probability is linear:

$$P(\text{reach } 4 \mid \text{start } k) = \frac{k}{4} \;\Rightarrow\; \boxed{\tfrac{1}{4}} \text{ from \$1.}$$

(Numerically, P^{500} has first interior row $(0.75, 0, 0, 0, 0.25)$.)
★ *Intuition:* A drunk on a plank of length 4 with cliffs at both ends: by symmetry the chance of falling off the far end equals the fraction of the plank you've already covered.

Problem B3 — Transient: the return probability is below 1

Chain on $\{A, B\}$: $P_{AA} = 0.5$, $P_{AB} = 0.5$, B absorbing. Find the probability of ever returning to A and the expected number of visits to A (including time 0).
Solution. The only way to return to A is to stay there one step: $f_{AA} = 0.5 < 1$, so A is **transient**. Visits to A form a geometric count:

$$\mathbb{E}[\text{visits}] = \frac{1}{1 - f_{AA}} = \boxed{2}.$$

★ *Intuition:* Each "stay" is a coin flip to keep the lights on; the first tail cuts them off forever. A recurrent state is a place you are *certain* to come back to (infinitely often); a transient state is a place you visit only finitely many times, with an expected count of $\frac{1}{1-f}$.
▶ *Speed trick:* Criterion: state i is recurrent $\iff f_{ii} = 1 \iff \sum_n P^n_{ii} = \infty$. Transient means this sum is finite.

Problem B4 — Positive vs. null recurrent vs. transient

A walk on $\{0, 1, 2, \dots\}$ steps right with probability $p = 0.3$, left with $q = 0.7$, and at 0 stays with probability 0.7. Classify it, find π_0 and the mean return time to 0. What happens when $p = q$ and when $p > q$?
Solution. With $\rho = p/q = \frac{3}{7} < 1$ detailed balance gives $\pi_k = (1-\rho)\rho^k$:

$$\pi_0 = \frac{4}{7}, \qquad \mathbb{E}[\text{return time to 0}] = \frac{1}{\pi_0} = \boxed{\tfrac{7}{4}}.$$

Finite expected return time ⇒ **positive recurrent**. When $p = q$ the walk still returns with probability 1 but the expected return time is infinite: **null recurrent** (no stationary distribution). When $p > q$ it drifts away: **transient**.
★ *Intuition:* A shopper tugged toward the front door. With a pull toward the door ($p < q$) you hover near it and return quickly. With no pull ($p = q$) you wander arbitrarily far before coming back — you *do* come back, but the average trip is infinite. With a pull outward ($p > q$) you may never return.

Problem B5 — Mean return time equals $1/\pi_i$

For A1's chain ($\pi = (\frac{2}{3}, \frac{1}{3})$) find the expected number of days to return to Sunny.
Solution. Formula: $m_S = 1/\pi_S = \boxed{1.5}$ days. First-step check: leave S for R w.p. 0.2 (then wait Geometric(0.4), mean 2.5 days to reach S):

$$m_S = 1 + 0.2(2.5) = 1.5. \;\checkmark$$

★ *Intuition:* If sunny days make up $\frac{2}{3}$ of all days in the long run, they occur on average every $\frac{3}{2}$ days. The long-run fraction of time in a state is the *reciprocal* of the mean gap between visits — exactly like bus frequency and bus interval.

C. Periodicity & Ergodicity

Problem C1 — Reading the period

(a) A walker on a 4-cycle moves $\pm 1 \pmod 4$ with probability $\frac{1}{2}$ each. (b) A deterministic loop $0 \to 1 \to 2 \to 0$. Find each period.
Solution. The period of i is $\gcd\{n : P^n_{ii} > 0\}$. (a) Returns happen at times $2, 4, 6, \dots$ ⇒ gcd = $\boxed{2}$. (b) Returns happen at times $3, 6, 9, \dots$ ⇒ gcd = $\boxed{3}$.
★ *Intuition:* Periodicity is a heartbeat: the chain can only be home at multiples of the period. A 4-cycle walk is like stepping square by square on a chessboard: every step flips the square's colour, so you can only be back on your starting colour after an even number of steps.

Problem C2 — A self-loop breaks periodicity: ergodic chains

$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Find the period, decide whether the chain is ergodic, and find π.
Solution. Cycles through state 0: the self-loop (length 1) and $0 \to 1 \to 2 \to 0$ (length 3): $\gcd(1, 3) = \boxed{1}$, so **aperiodic**. The chain is irreducible and finite, hence positive recurrent: **ergodic** (= irreducible + aperiodic + positive recurrent). Balance:

$$\pi_1 = 0.5\pi_0, \quad \pi_2 = \pi_1 \Rightarrow \pi = \boxed{(\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}.$$

★ *Intuition:* One "wait a beat" loop is enough to scramble the rhythm: once you can return in 1 step and in 3 steps, you can return in 4, 5, 6, …, i.e. at *every* large time. The chain stops dancing to a metronome and starts to settle.

Problem C3 — Convergence of P^n (and when it fails)

For A1's chain, compute P^5, P^{10} and describe the rate. Contrast with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
Solution.

$$P^5 = \begin{pmatrix} 0.670 & 0.330 \\ 0.660 & 0.340 \end{pmatrix}, \qquad P^{10} = \begin{pmatrix} 0.6667 & 0.3333 \\ 0.6666 & 0.3334 \end{pmatrix}.$$

Rows converge to $\pi = (\frac{2}{3}, \frac{1}{3})$ geometrically at rate $|\lambda_2|^n = 0.4^n$ (the second eigenvalue is 0.4). For the swap chain, P^n alternates between I and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and **never converges**, although $\pi = (\frac{1}{2}, \frac{1}{2})$ still solves $\pi P = \pi$.
★ *Intuition:* Ergodic chains forget their past: after about 10 days the weather today tells you nothing about the weather next week. A periodic chain never forgets — it remembers the *parity* of the clock forever.
▶ *Speed trick:* Convergence time $\approx \ln(\varepsilon)/\ln|\lambda_2|$: gaps near 1 mean slow mixing.

D. Stationary Distributions

Problem D1 — Two-state formula

For A1's chain ($P_{SR} = 0.2$, $P_{RS} = 0.4$), find the stationary distribution.
Solution. Balance of flow: $\pi_S(0.2) = \pi_R(0.4)$ and $\pi_S + \pi_R = 1$:

$$\pi = \left(\frac{0.4}{0.2 + 0.4}, \frac{0.2}{0.2 + 0.4}\right) = \boxed{(\tfrac{2}{3}, \tfrac{1}{3})}.$$

★ *Intuition:* Two rooms connected by doors. In equilibrium, as many people walk S→R per minute as R→S. The room with

the leakier exit holds fewer people: $\pi_S/\pi_R = P_{RS}/P_{SR}$.

▶ *Speed trick:* Two states: $\pi = \left(\frac{q}{p+q}, \frac{p}{p+q}\right)$ where $p = P_{12}$, $q = P_{21}$.

Problem D2 — Solving $\pi = \pi P$ for three states

For A4's market chain, find π.

Solution. The equations (column balance) are

$$0.3\pi_B = 0.3\pi_F + 0.2\pi_R,$$

$$0.6\pi_F = 0.2\pi_B + 0.3\pi_R,$$

$$0.5\pi_R = 0.1\pi_B + 0.3\pi_F.$$

From the third, $\pi_R = 0.2\pi_B + 0.6\pi_F$. Substitute into the first: $0.26\pi_B = 0.42\pi_F \Rightarrow \pi_F = \frac{13}{21}\pi_B$, $\pi_R = \frac{12}{21}\pi_B$. Normalizing,

$$\pi = \left(\tfrac{21}{46}, \tfrac{13}{46}, \tfrac{12}{46}\right) \approx \boxed{(0.457,\ 0.283,\ 0.261)}.$$

★ *Intuition:* Think of probability as sand poured over three buckets. Every day each bucket sends grains along the arrows. The stationary pile sizes are the ones where each bucket's outflow exactly equals its inflow — the level stops changing even though sand keeps moving.

▶ *Speed trick:* Use $n-1$ balance equations plus $\sum \pi_i = 1$ (the n balance equations are redundant). Check your answer with the equation you did *not* use.

Problem D3 — Birth–death chains and detailed balance

A walk on $\{0,1,2,3\}$ moves right w.p. 0.3, left w.p. 0.7, with staying at the ends (stay w.p. 0.7 at 0, 0.3 at 3). Find π.

Solution. For nearest-neighbour chains, detailed balance $\pi_k P_{k,k+1} = \pi_{k+1} P_{k+1,k}$ gives $\pi_{k+1} = \frac{3}{7}\pi_k$. Weights $1, \frac{3}{7}, \frac{9}{49}, \frac{27}{343}$ sum to $\frac{580}{343}$:

$$\pi = \frac{1}{580}(343,\ 147,\ 63,\ 27) \approx \boxed{(0.591,\ 0.253,\ 0.109,\ 0.047)}.$$

★ *Intuition:* On a line, flow across each *doorway* must balance separately — no need to solve a big linear system. It's like checking the water level in a chain of tanks connected by pipes: equal flow each way across every pipe means the whole system is still.

Problem D4 — Doubly stochastic ⇒ uniform

$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.5 & 0.2 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{pmatrix}$. Find π without solving anything.

Solution. Rows *and* columns sum to 1 (a doubly stochastic matrix). Then the uniform vector is stationary:

$$\left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right) P = \left(\tfrac{1}{3}(0.2+0.5+0.3), \ldots\right) = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right).$$

Since the chain is irreducible and aperiodic, $\boxed{\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)}$ is the unique answer.

★ *Intuition:* If every state is equally "hard to leave" *and* equally "easy to enter," nothing favours one state over another — a perfectly fair carousel.

▶ *Speed trick:* Always check column sums first: doubly stochastic means uniform, in $O(n^2)$ time instead of $O(n^3)$.

Problem D5 — Random walk on a graph

A random walker on the graph with edges 1–2, 2–3, 3–4, 4–1, 1–3 moves to a uniformly random neighbour. Find π.

Solution. Degrees: $d = (3,2,3,2)$ (total 10). Stationary probability is proportional to degree:

$$\pi_i = \frac{d_i}{\sum_j d_j} = \boxed{(0.3,\ 0.2,\ 0.3,\ 0.2)}.$$

Check detailed balance on edge 1–2: $\pi_1 \cdot \frac{1}{3} = 0.3 \cdot \frac{1}{3} = 0.1$ and $\pi_2 \cdot \frac{1}{2} = 0.2 \cdot \frac{1}{2} = 0.1$. ✓

★ *Intuition:* A well-connected intersection gets more traffic. In a random walk on a network, the long-run time at a node is proportional to the number of roads in — exactly the idea behind PageRank (with teleportation added so the chain is ergodic).

E. Quant Capstones

Problem E1 — Long-run return of a regime-switching strategy

Daily returns by regime are +2% (Bull), 0% (Flat), −3% (Bear). With A4's chain, find the long-run average daily return.

Solution. By the ergodic theorem, the long-run time average equals the stationary average:

$$\bar{r} = \sum_i \pi_i r_i = \frac{21}{46}(2\%) + \frac{13}{46}(0) + \frac{12}{46}(-3\%)$$

$$= \frac{42-36}{46}\% = \boxed{\approx 0.13\% \text{ per day}}.$$

Over 252 trading days this is about 0.33 in cumulative simple return.

★ *Intuition:* The bull regime's big gains are mostly cancelled by the bear regime's big losses; what is left is a thin edge of 0.13% per day. Without computing π, the answer would look like "Bull is common, so the strategy is great" — the Markov structure corrects the intuition.

Problem E2 — Expected flips until HH (a Markov chain in disguise)

Flip a fair coin until two consecutive heads appear. Find the expected number of flips.

Solution. States: 0 (no progress / last was T), H (last flip was H). With e_s the expected remaining flips:

$$e_0 = 1 + \tfrac{1}{2}e_H + \tfrac{1}{2}e_0, \qquad e_H = 1 + \tfrac{1}{2}(0) + \tfrac{1}{2}e_0.$$

Substituting: $e_0 = 2 + e_H$ and $e_H = 1 + \frac{1}{2}e_0 \Rightarrow e_0 = \boxed{6}$.

★ *Intuition:* "Memory" for a pattern problem is just the longest useful suffix of what you've seen, so a coin sequence becomes a tiny Markov chain. HH takes 6 flips on average, while HT takes only 4 — after a failed HH attempt you slide back to the start, but after a failed HT attempt you are already halfway there.

▶ *Speed trick:* Pattern waiting-time problems ⇒ build the suffix-state chain, write one first-step equation per state, solve the small linear system.

Drill order for review: re-derive B1 (absorption probabilities by first-step analysis), B4 (positive vs. null recurrence), D2 (solving $\pi = \pi P$) and E2 (pattern chains) from scratch, closed-book. They carry the chapter's central ideas: classify first (absorbing/transient/recurrent), check periodicity, then solve for π only when the chain is ergodic.